ST Engineering InnoChamp 2026 (internal innovation programme)

INVESTOR PITCH · [DATE]

AGIL® Secure FastPass

Departure capacity without building a terminal.

[Name], Chief Product Officer · [Name], Chief Marketing Officer

Asking [US$X M] in four tranches · lane live [Qx 2027] · paid order [Qx 2028]

Problem

Up to four ID checks per departure, most staffed.

Airports are running out of space; border agencies, out of people.

1 Check-in | 2 Immigration | 3 Security screening | 4 Boarding gate



AIRPORT

~1.5×

Manila: 52M passengers through terminals built for ~35M

8–10 yrs

to add a terminal: Changi T5 and Dubai DWC

BORDER AGENCY

"ICA's manpower cannot keep growing indefinitely."

Singapore Home Affairs Minister, 2025

GMA News (Jan 2026) · Singapore MOT (Jan 2025) · Dubai Media Office (Apr 2024) & WAM (Jun 2026) · MHA, ICA Workplan Seminar 2025 · Staff count: internal estimate

Why now

Borders went biometric; passports are going mobile.

Buyers, travellers and standards moved in 2024–26.

54% today

→ **83%**

surveyed airports with biometric borders (2028, expected)

79%

of passengers surveyed ready for a phone passport

~127M

passport-less clearances in Singapore, 2025

Sep 2024 · Changi passport-less, all terminals

Nov 2025 · Indonesia walk-through corridors

Apr 2026 · ICAO drafts phone-credential standards

2027 · FastPass pilot (target)

SITA Air Transport IT Insights 2025 (Apr 2026) · SITA Passenger IT Insights 2025 (Oct 2025) · ICA (Oct 2024; 2025 statistics) · Amadeus (Nov 2025) · ICAO FALP/14-WP/21 (proposed, not adopted)

Our insight

Phone in the pocket + face in stride = cleared.

Corridors match a stored face. We add the phone you carry.



Device credential **Live face** **Cleared**

Up to 4 → 0

document stops at full rollout (target)
pilot lane: 1 → 0

Two factors by design; independent tests at M2.

Our bet, tested at M1: a second factor opens walk-through to more travellers.

Walk-through example: Dubai Red Carpet corridor, pre-registered, face only (GDRFA via International Airport Review, 2026) · Eligibility limits today: ICA (Oct 2024)

Product walkthrough

One traveller, four checks, no documents shown.

Illustrated walkthrough of the prototype shown at Singapore Airshow, Feb 2026.

- **Check-in** → enrol in app [+ airline online check-in]
- **Immigration** → Bluetooth (BLE) lane + e-gate
- **Security ID** → Nova Plus reader; bags still X-rayed
- **Boarding** → kiosk: face + phone

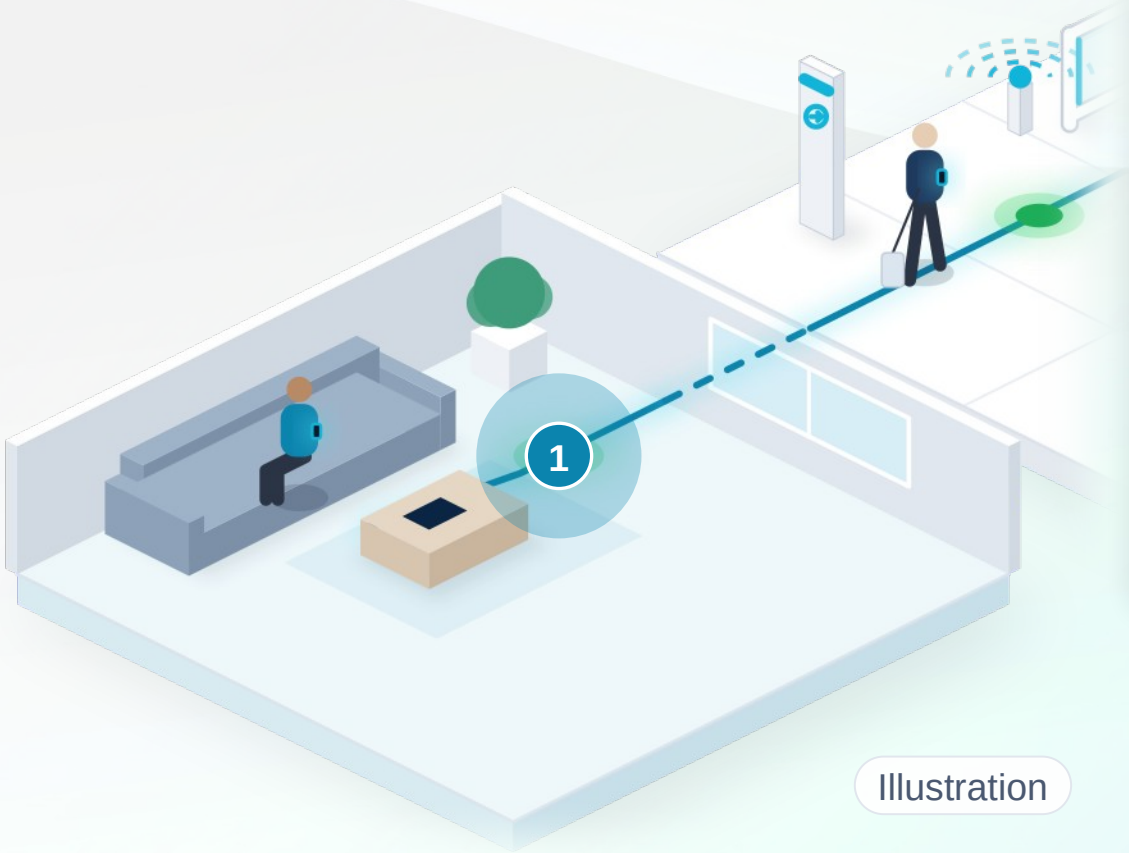


Illustration

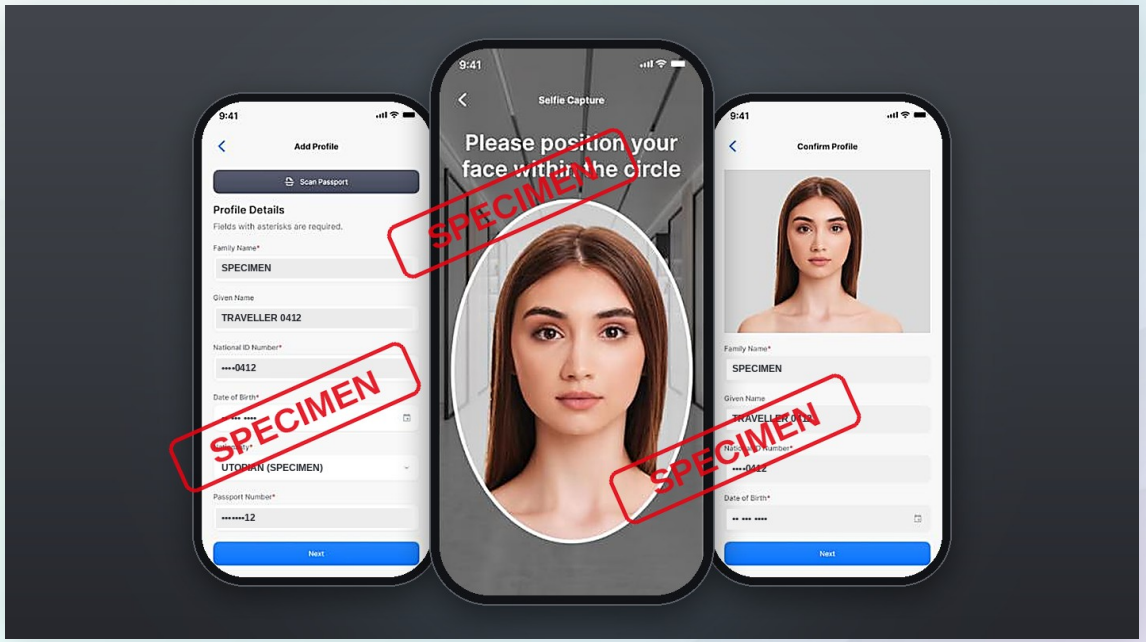
Illustration of the prototype; controlled demo conditions. Eligibility set by each border agency. Traveller data synthetic; app screens are specimens.

Product walkthrough

1 · Enrol once, at home



Illustration

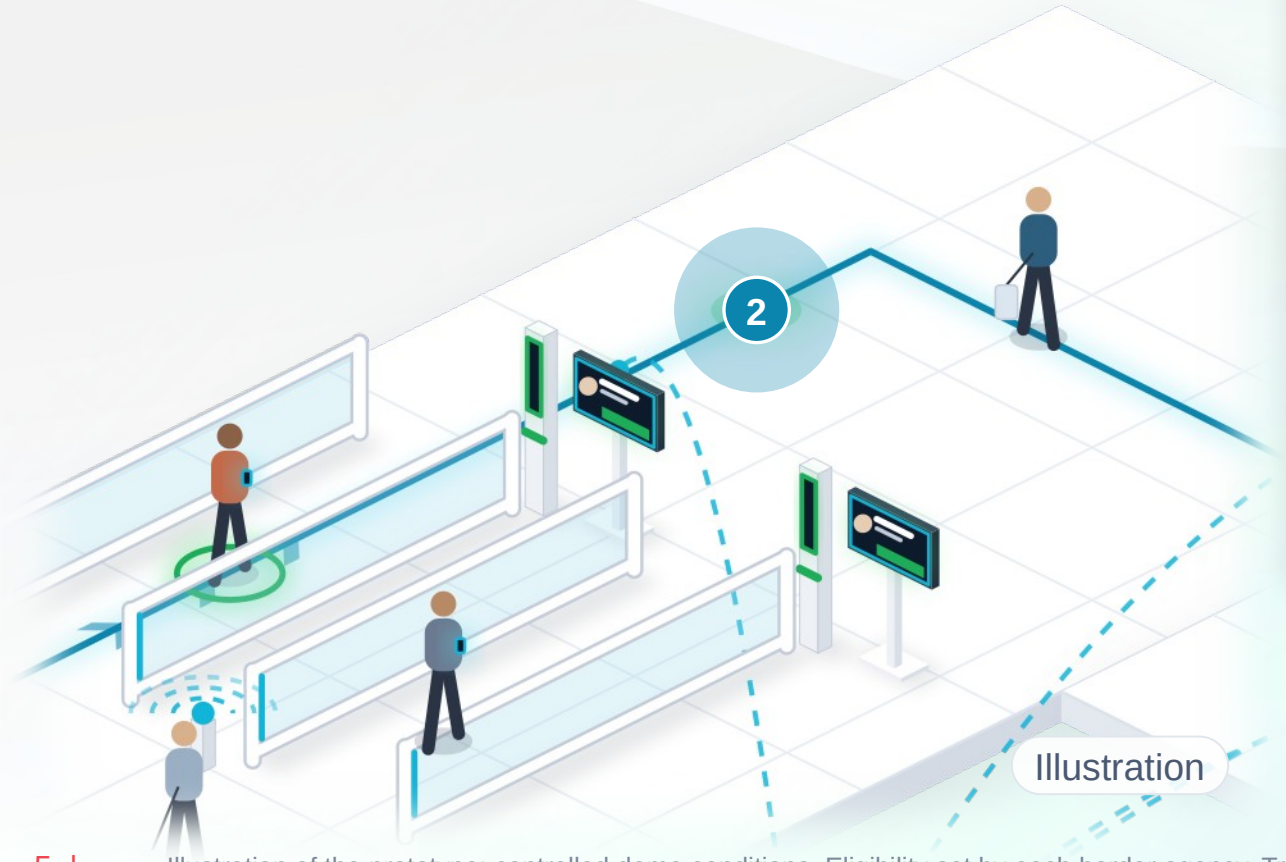


Passport scan and selfie, once. Encrypted on the phone; SDK for airline apps.

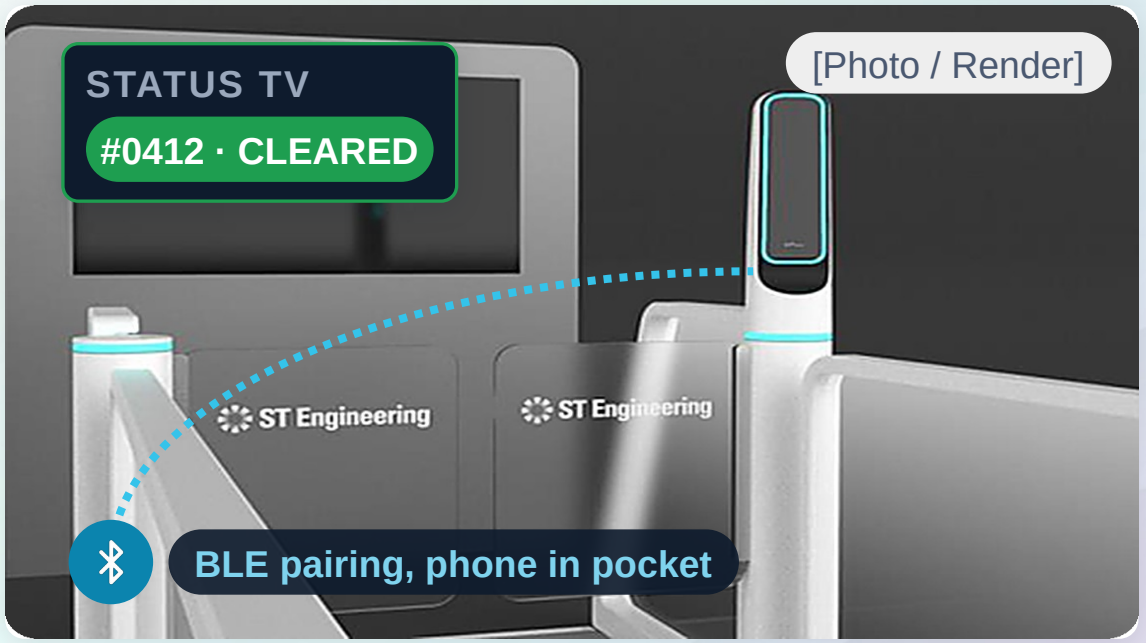
Illustration of the prototype; controlled demo conditions. Eligibility set by each border agency. Traveller data synthetic; app screens are specimens.

Product walkthrough

2 · Walk through: phone + face = cleared



Illustration

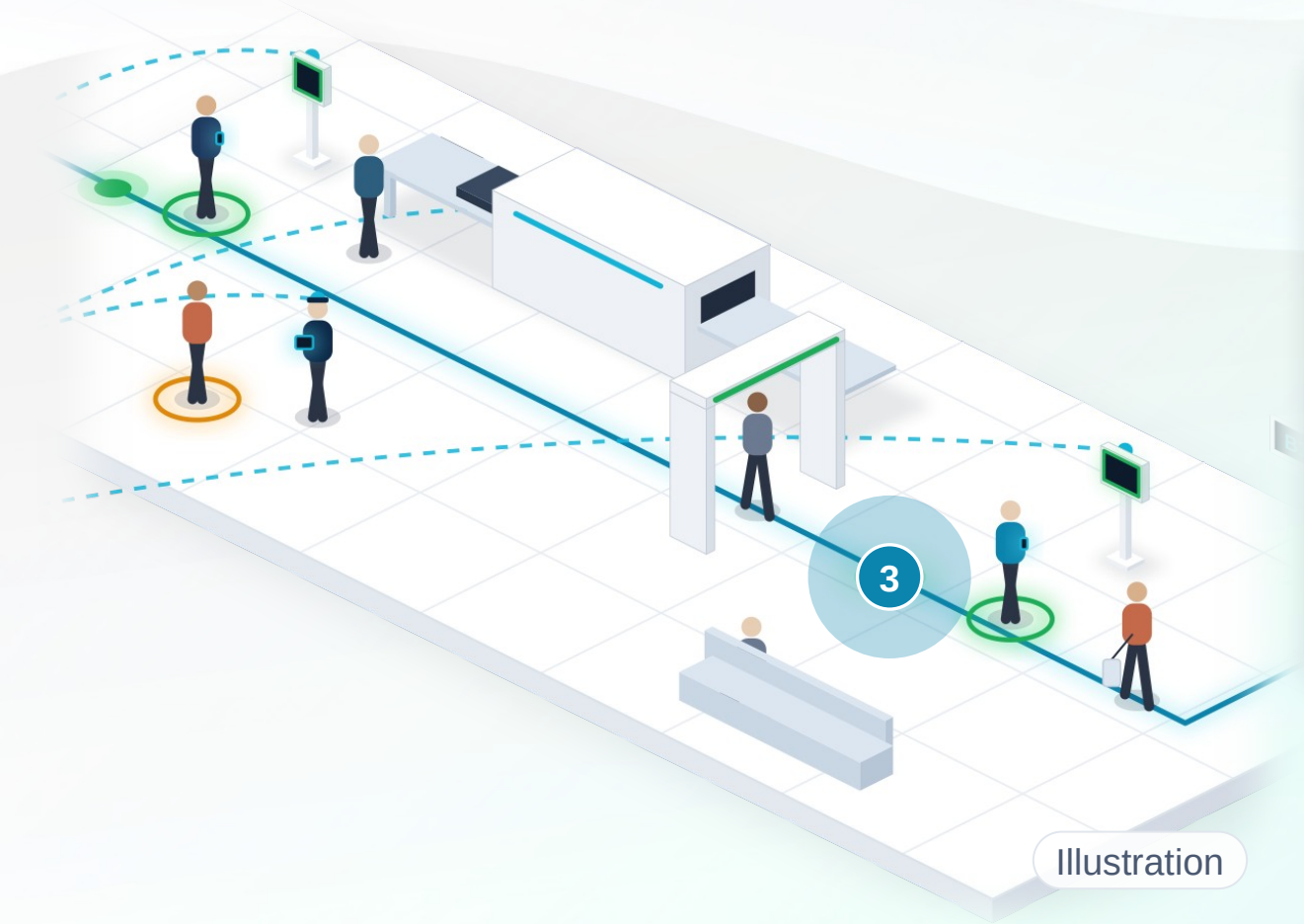


[X] s per traveller, prototype, controlled conditions [measure or delete]

Illustration of the prototype; controlled demo conditions. Eligibility set by each border agency. Traveller data synthetic; app screens are specimens.

Product walkthrough

3 · Pre-screen and board by face + phone



Illustration



Nova Plus re-confirms identity before X-ray. Boarding needs airline departure-control integration.

Illustration of the prototype; controlled demo conditions. Eligibility set by each border agency. Traveller data synthetic; app screens are specimens.

4 · Officers act only on exceptions



Illustration

Time	Traveller	ID	Lane	Status
10:12:41	TRAVELLER 0413	••••0413	1	EXCEPTION
	No BLE credential	Officer review		*mock-up*
10:12:34	TRAVELLER 0412	••••0412	1	CLEARED
10:12:31	TRAVELLER 0411	••••0411	2	CLEARED
10:12:27	TRAVELLER 0410	••••0410	1	CLEARED
10:12:22	TRAVELLER 0409	••••0409	2	CLEARED
10:12:18	TRAVELLER 0408	••••0408	1	CLEARED

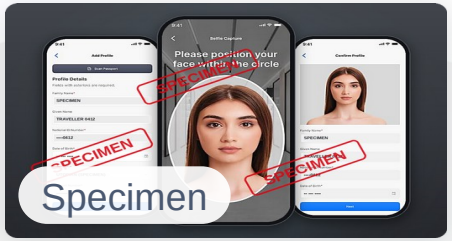
Clearance events sit beside CCTV, access control and incidents.

Illustration of the prototype; controlled demo conditions. Eligibility set by each border agency. Traveller data synthetic; app screens are specimens.

Traction & learning

Built and shown live. Next proof: one airport lane.

Pre-revenue; no pilot or letter of intent signed yet.



App



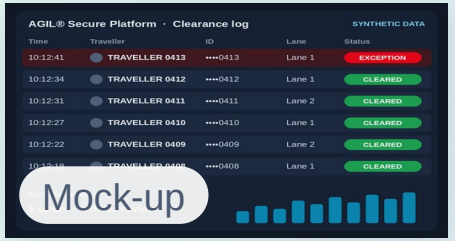
E-gate + BLE



Nova Plus



Kiosk



Platform

[N]

airports and border agencies in pilot talks

Pilot scorecard, agreed before go-live

- Throughput vs e-gate
- Officer minutes
- False accepts / rejects
- Enrolment
- Kerb-to-gate (full-rollout target −20–30%)

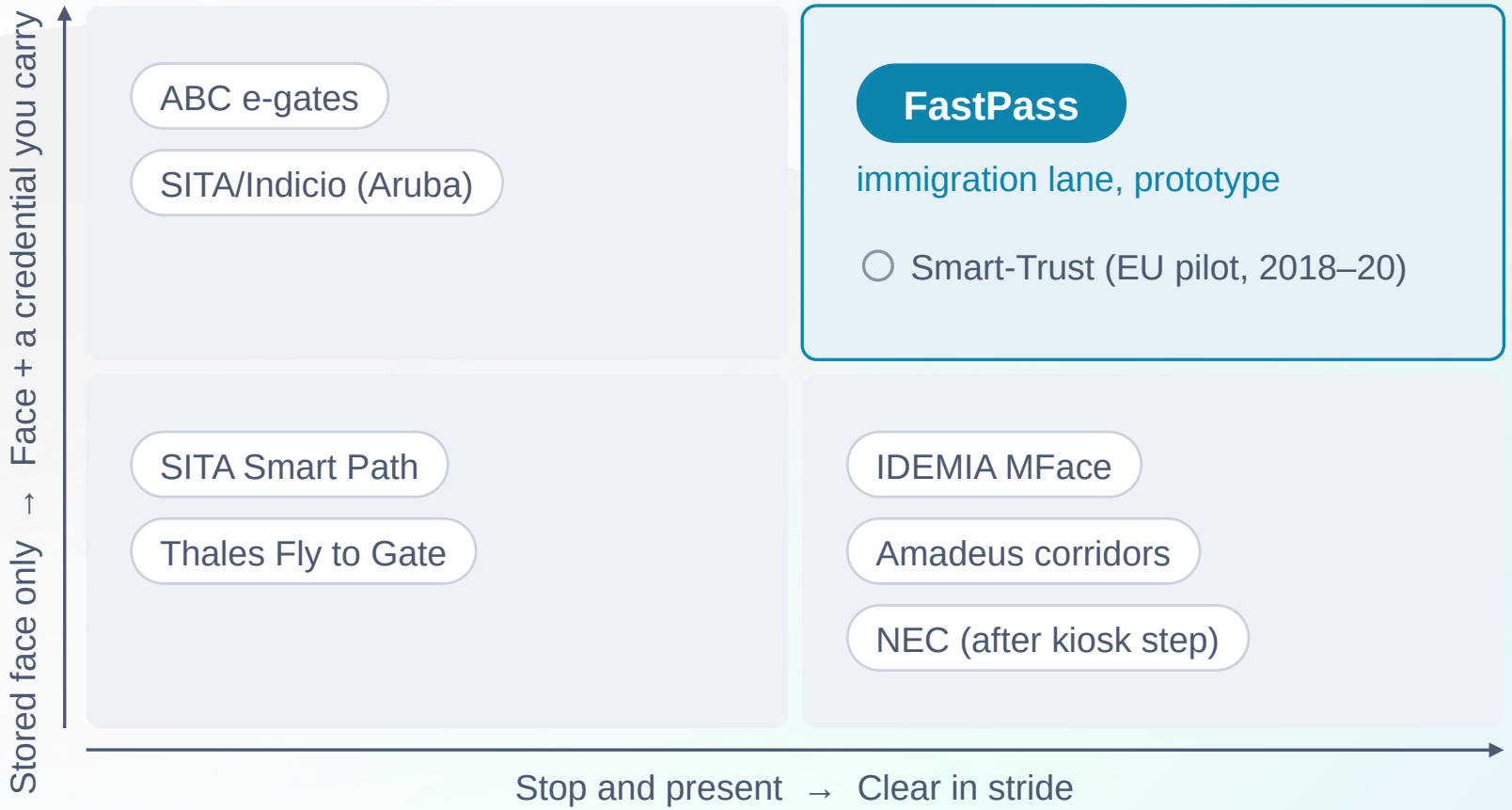
Staffed fallback lane and officer override

Tiles: product images [photo / render, confirm], specimen and mock-up screens · Company record: Singapore Airshow, Feb 2026 · Full scorecard, baseline and go/no-go: A7

Competition

Others match faces in stride. We add the phone.

Our wedge: two factors, nothing taken out, plus the security control room.



Our wedge

(to prove in the pilot)

- ✓ Two factors, nothing taken out, in stride
- ✓ Clearance events in the security control room
- ✓ Starts with one lane in an existing terminal

€320M

Amadeus–Vision-Box EV, Apr 2024: a comp, not a valuation

Placement from public sources; 'not found' ≠ 'cannot'. IDEMIA · SITA · PhocusWire · Singapore EDB · Amadeus · NEC · CORDIS 778571 · Detail: A4

Market

US$42–94M a year by 2030, and lanes are only the wedge.

Deliberately narrow: e-gate and kiosk spend only, two regions.

US$4.18B airport ABC market, 2030

Not to scale

× 10–15% Middle East + SE Asia **est.**

US$0.42–0.63B

× 10–15% FastPass-serviceable **est.**

US$42–94M ≈ 1–2.25% of the airport market

Restated: deck 12 read 'US$50–75M over 5 years'; the same inputs give US$42–94M a year.

target

Revenue target: US$10–15M a year by [year], ≈15–23% of the ~US$65M midpoint (target, to confirm); ≈40–100 gate-equivalents a year at tender prices

Where it grows (hypothesis)

ME + SEA lanes → **Every touchpoint** → **Airports worldwide** → **Other borders [confirm]**

Per year, 2030: US$5.03B × 83.1% = US$4.18B; × 10–15% × 10–15% = US$42–94M (midpoint ~US$65M at 12.5% × 12.5%). Shares: internal estimates (URS).
ABC = e-gates and kiosks, mostly one-off hardware; 83.1% is the 2025 airport share applied to 2030. MarketsandMarkets, Automated Border Control Market to 2030.

Business model

Land one lane. Hardware once, software every year.

One standard product core, reused across airports, is how margins grow.

1 — 2 — 3 — 4

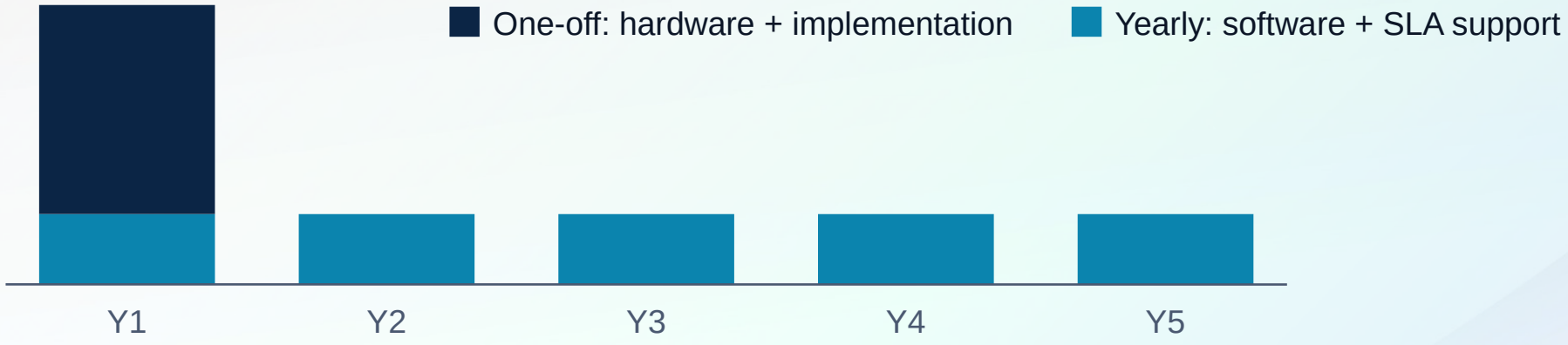
Pilot lane
1 lane, 1 group
[paid / co-funded]

Terminal
more lanes + platform licence

Airport
every touchpoint, one platform

Portfolio
more airports, one core

Revenue per lane (illustrative, not a forecast)



Revenue lines from the team's commercial model · Tender price anchors (Riga, Aena) and margin comparables: A11

Buyer case & go-to-market

Airport buys capacity. Agency gains officer time.

Payback is agreed per payer, before go-live.

BUYS **APPROVES**

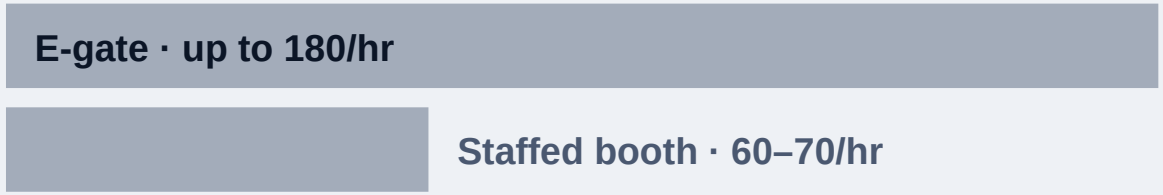
Airport operator
more departures per lane

Border agency
officers on exceptions only

Airline
enrolment in its app

Traveller
enrols once, no documents

E-gate benchmark, not FastPass



1 officer : 4–10 e-gates

FastPass: [X]/hr target, vs this baseline

Riga Airport / State Border Guard of Latvia (AviationPros) · UK Border Force Aviation Forum minutes (Feb 2018) · Payback per payer: A11 · Target filter: A3

Team

The team that built and demoed the prototype.

Three ST Engineering business areas; venture-built via STEV / In.Vent [structure TBC].

[Name]
◌ [FT / PT]
[Venture lead / GM]
[proof, ≤5 words]

[Name]
◌ [FT / PT]
Product (CPO)
[proof]

[Name]
◌ [FT / PT]
Marketing (CMO)
[proof]

[Name]
◌ [FT / PT]
[Technology: biometrics, BLE, platform]
[proof]

Open role
Hiring now:
commercial lead
+ certification & privacy lead (programme)

Advisors
[border-agency official]
[airport operations executive]

● full-time ○ part-time

Why ST Engineering, why a venture

- 35+ years of secure delivery (company claim)
- Group pull-through [US$ target, TBC]
- Structure and IP owner [TBC]
- Why a venture, not a feature: [rationale, TBC]

THE ASK

[US$X M] in four tranches; first lane live by [Qx 2027].

Four milestones, four tranches: no milestone met, no next tranche.

M1 · [Qx 2027]
T1 · [US$A M]
Pilot signed: airport + agency
Risk retired: **agency**

M2 · [Qx 2027]
T2 · [US$B M]
Lane live, independent tests passed
Risk retired: **technical**

M3 · [Qx 2028]
T3 · [US$C M]
KPIs read out; miss → stop or redesign
Risk retired: **operational**

M4 · [Qx 2028]
T4 · [US$D M]
First paid multi-lane order
Risk retired: **commercial**

Today we ask for

1 [Approval of the programme and Tranche 1]
2 [One introduction to a pilot airport and its border agency]
3 [Name an executive sponsor]

The trusted identity layer for every checkpoint.

 [Name, role, email]

Appendix

Appendix: backup for diligence and Q&A

Click a row to open it. Every external figure is mapped to a source in A1.

A1	Source map	**A9**	Regulatory and standards path
A2	Full journey, hardware and the BLE lane	**A10**	Risks and the milestone that retires each
A3	Market maths, segment and regional context	**A11**	Unit economics, payback per payer and plan
A4	Competitor matrix, evidence and moat	**A12**	Team detail and what an investor would own
A5	Category exits and consolidation	**A13**	Assumptions register
A6	AGIL® Secure Platform detail	**A14**	Q&A prep: ten killer questions
A7	Pilot design and full scorecard	**A15**	Roadmap and vision options
A8	Trust, privacy and security	**A16**	Ship gates and fill list (INTERNAL)

Back to the ask

Every external number maps to one source.

Short tags on slides; this slide's notes list all 78 sources (id · title · URL).

Main deck

- S2 · DEM11, DEM10, FAC14, DEM9, DEM9B, DEM13
- S3 · FAC4, WHY10, WHY12, WHY11, COM2, WHY6
- S4 · WHY13, WHY11
- S7 · COM5, COM2, COM9, COM8, COM7, COM10, COM3, COM1, COM14, WHY14
- S8 · FAC6, FAC7, FAC8, DEM2
- S9 · ROI8, ROI9
- S10 · ROI5, ROI6, FAC13

Appendix and Q&A

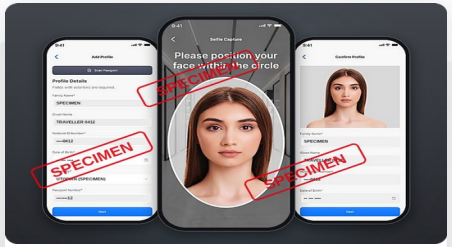
- A3 · FAC5, FAC6–FAC9, FAC10/DEM6, DEM2–DEM5, DEM8, DEM9, DEM9B, DEM11, DEM12, DEM12B
- A4 · COM1–COM12, COM5B, COM14, FAC12, FAC15, WHY11, WHY13, WHY13B, WHY14
- A5 · COM1, COM13, COM14, COM14B, COM14C, COM15, COM15B
- A7 · ROI2, ROI4, ROI5 | A8 · WHY9
- A9 · WHY1, WHY1B, WHY2–WHY8, WHY15, WHY16
- A10 · DEM2–DEM4, DEM9, DEM9B, FAC4, WHY2, WHY7, WHY14, FAC13
- A11 · FAC1, FAC2, ROI1, ROI2, ROI4, ROI5, ROI8, ROI9, ROI11–ROI16, ROI14B
- A13 · DEM6 | A14 · DEM2, DEM9, DEM9B, WHY14, ROI8, ROI9, ROI11

G-SRC gate · [CMO] · due [date]: click-check every URL and wording before either file leaves the team; unconfirmed → delete or mark '(reported by [outlet])'; conflicts → conservative value; also check WHY1B and ROI14B (added from memory). Research opened no primary page.

Appendix index

Six steps, five modules, one platform.

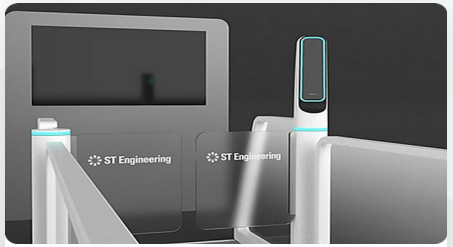
Backup for CTO and COO product questions. Live status per module to confirm.



1 · Enrol

FastPass Mobile App

[Shown live / concept]



2–3 · Approach, walk

BLE lane + biometric e-gate

[Shown live / concept]



4 · Pre-screen

Nova Plus check; X-ray still applies

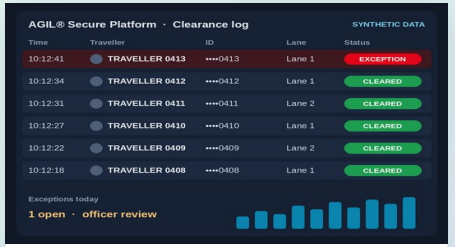
[Shown live / concept]



5 · Board

Smart Boarding Kiosk

[Shown live / concept]



6 · Oversee

AGIL® Secure Platform

[Shown live / concept]

Claim fix

- No documents shown at checkpoints for enrolled, eligible travellers
- The passport is still carried as required (ICAO DTC Type 1)

Check-in coverage

- No module is a check-in or bag-drop device
- Check-in = app enrolment + airline online check-in [confirm]; bag drop out of scope

Counts reconciled

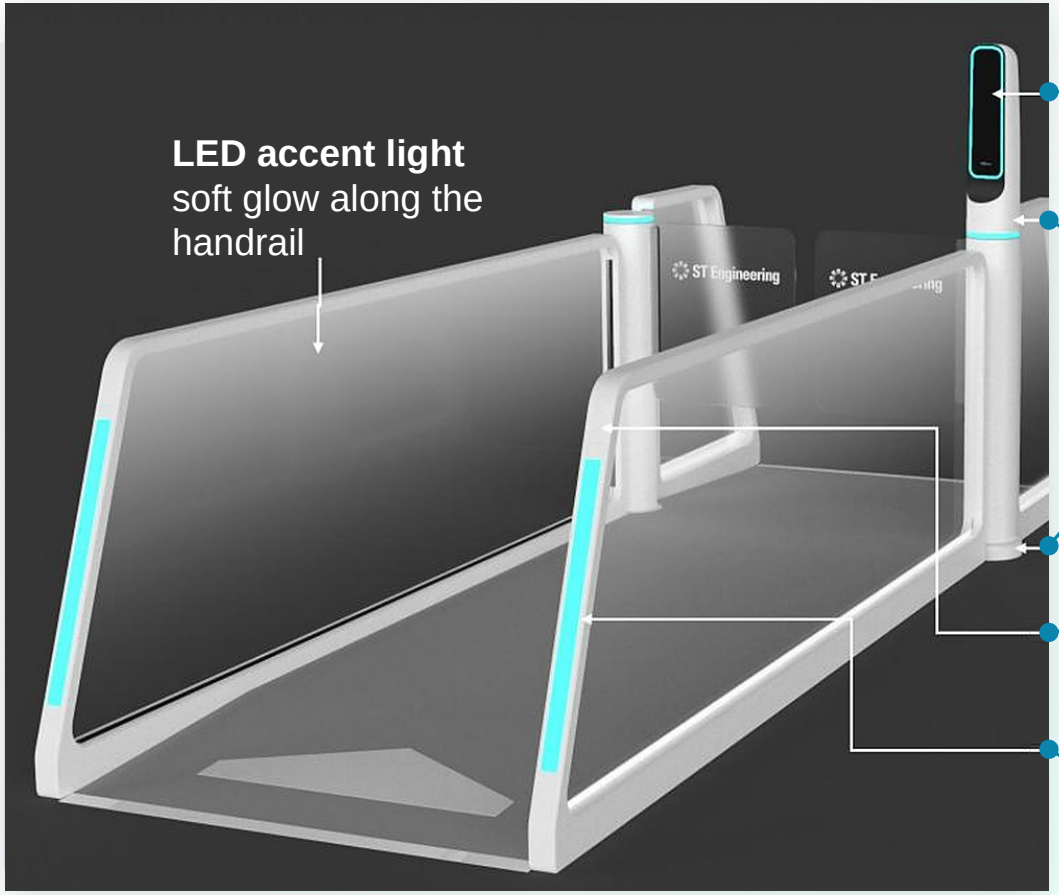
- 4 checks · 5 modules · 6 steps · 1 platform
- Walkthrough 1–4 = steps 1 · 2–3 · 4–5 · 6
- AI luggage scanning: roadmap (A15)

All demo screens show specimen or synthetic data · Image types [photo / render] to confirm

Appendix index

The FastPass lane: BLE beacon plus a face camera.

Prototype lane hardware. Image type [photo / render] to confirm.



FR pylon gantry
Pylon, G/R/B lights and face camera

Flexible mounting angle
Adjusts to gate layouts and user heights

Pylon mounted on the floor
Permanent setup for stability

BLE beacon
Pairs with the phone; proximity detection

Indicator light
Colour-coded system status

Design intent [CTO to confirm]

- The lane challenges the credential on the paired phone, not just a beacon.
- The live face must match that credential.
- Wrong pairing → exception to an officer, never a false accept.
- Officers see the exception in AGIL® Secure Platform.

Recompute the beachhead: US$42–94M a year in 2030.

Every input and step, plus what the figure does not cover.

Funnel, per year in 2030

- US$5,030M × 83.1% = US$4,180M
- × 10–15% = US$418–627M
- × 10–15% = US$41.8–94.0M
- Midpoint at 12.5% × 12.5% = US$65.3M (mean of the range: US$68M)
- To scale: 0.8–1.9% of global ABC, 1.0–2.25% of airport ABC (hence 'Not to scale')

Other readings

- Cumulative 2026–30 (US$19.5B path): US$162–365M
- 2025 base (US$2.56B): US$21–48M a year
- Each 1-point change in either share: ≈ US$4.2–6.3M a year
- Independent estimates: ABC US$5.0–5.4B by 2030 (low confidence)

Input labels

- 83.1% = 2025 airport share, applied to 2030
- Asia-Pacific 32.6% (2024) includes China, Japan, India: not a proxy for SEA

Scope caveat

- ABC = border e-gates and kiosks, mostly one-off hardware
- FastPass software, check-in, pre-screen, boarding and the platform sit outside it
- So the beachhead understates scope, but it is not a software market

Bottom-up, to build before external use

- [N] airports × [N] lanes × US$[X] one-off + US$[Y]/yr = US$[Z]M a year; must reconcile with the top-down

MarketsandMarkets, Automated Border Control Market to 2030 · Research and Markets / Mordor; Lucintel · Shares: internal estimates (URS)

Appendix index

Who we target first, and the 2026 headwind.

Proposed segment for the team to confirm; regional data for the pilot-market call.

Segment (proposed)

- 'Congested international gateways that cannot add a terminal soon'
- Filter: brownfield · at or above design capacity at peak · no incumbent end-to-end biometric programme (not Changi, DXB, Doha) · Group relationship · regulator open to pilots
- Test NAIA-type gateways (~50M passengers). First three: [A] · [B] · [C]

Brownfield

- ~78% of ME/SEA airport projects are brownfield: internal URS figure [state whether it counts projects]
- ACI APAC & MID: 57% of US$240B capex 2025–2035 goes to brownfield, by value
- Different metrics: never merge them

Wider pool, not TAM

- Airports spent US$14.8B on IT in 2025, 7.3% of revenue (SITA)

2026 regional context

- ME airport traffic −21% in H1 2026 (ACI); ME carriers −14.2% YoY in Aug (IATA)
- IATA: ME demand −11.4% for 2026; ACI +5.4%/yr (2025–28) is pre-war
- Saudi: 140.9M passengers in 2025; Jeddah ~107% of design capacity [verify]
- Bangkok: 47% still need an officer; East Expansion slipped to 2030
- Dubai DWC phase 1 opens 2032

So

- Sell efficiency, not growth; SEA or dual-region pilot as the hedge

Competitor matrix behind the slide-7 map.

Public sources only; 'not found in public sources' ≠ 'cannot'.

Vendor	Face in stride	Credential in lane	Stays in pocket	Ops platform	Notable deployments
IDEMIA	Yes (MFace)	Yes, e-gate (stop)	Not found [verify]	[verify]	Singapore ABCS (May 2023); Zayed >1M (vendor); ID2Travel app
SITA	No (stop-and-look)	Yes, Aruba (tap, stop)	No (tap)	[verify]	Smart Path on NEC at Frankfurt; Hamad 'Fast Pass' (Jul 2026)
Amadeus / Vision-Box	Yes (corridors)	Pilot only (Smart-Trust, 2018–20)	[verify]	Yes (Orchestra)	Indonesia: >30/min per corridor (vendor); DXB T3 (2020); Smart-Trust EU pilot
NEC	Yes (after kiosk step)	None found	No	[verify]	Japan ISA: Kansai, Narita, Haneda (Apr 2025)
Thales	No (pod, stop)	Not in lane [verify]	No [verify]	[verify]	Fly to Gate: 600+ touchpoints, 7 Adani airports (vendor)
Collins · Indra	No (stop-and-look)	Yes, document (stop)	No	[verify]	SelfPass, Haneda (2021); Indra ABC at Aena (2018)
FastPass	**Yes (prototype)**	**Yes, phone credential**	**By design (BLE) [CTO]**	**Yes, AGIL® Secure Platform**	**Prototype shown Feb 2026; no deployments yet**

IDEMIA · SITA · PhocusWire · Amadeus · NEC · Singapore EDB (Thales) · Collins · Indra · CORDIS 778571 · Hamad International · ABI Research (Jul 2023)

Appendix index

Market evidence and the moat to build.

Agencies and airports are possible buyers, so they are not plotted as rivals.

Market evidence (not plotted)

- Dubai GDRFA Red Carpet: 745,188 travellers Jan–Aug 2026; 3.4 s vs 12.5 s (GDRFA; Paravision ~14 s); face only
- Hamad 'Fast Pass' (Qatar Airways + SITA): 700+ touchpoints, app enrolment
- Changi ICA passport-less: 25 s → 10 s; foreign visitors on departure only
- Never imply any ICA/HTX partnership
- Leaders by ranking: IDEMIA, Thales, Vision-Box (ABI Research, Jul 2023)

Moat to build (honest)

- Accreditation earned in the pilot
- Control-room integration: a switching cost only where security ops run on AGIL® Secure Platform
- Government-grade delivery in ME/SEA
- IP: [state the actual status]
- Copy in 12 months? [team answer]: copy risk is real

Wording rules (INTERNAL)

- Never 'first', 'only' or 'unique'
- 'Not found' ≠ 'cannot'
- ABC e-gates are two factors (chip + face), stop and present
- Wedge: 'two factors, nothing taken out, in stride'

Gulf News (2026) · International Airport Review (2026) · Paravision · Hamad International / SITA (Jul 2026) · ICA (Oct 2024) · ABI Research (Jul 2023)

Appendix index

The identity layer produces real outcomes.

Comps for context, not a valuation of FastPass.

Deal	Value	Detail	Date
Amadeus–Vision-Box	~€320M EV	~16× 2023 normalised EBITDA (per The Corner); ~€70M expected 2023 revenue (~4.6×, derived); ~470 staff	Apr 2024
BigBear.ai–Pangiam	~US$70M, all-stock	Announced Nov 2023	Early 2024
SITA–Materna IPS	Not disclosed	Airport solutions	2024
IN Groupe–IDEMIA Smart Identity	Not disclosed	—	Jul 2025
Thales–Gemalto	€4.8B	—	Apr 2019
Persona Series D	US$200M at US$2B	Co-led by Founders Fund and Ribbit	Apr 2025
CLEAR Secure IPO	US$409M raised at US$31	38 airports at IPO	Jun 2021

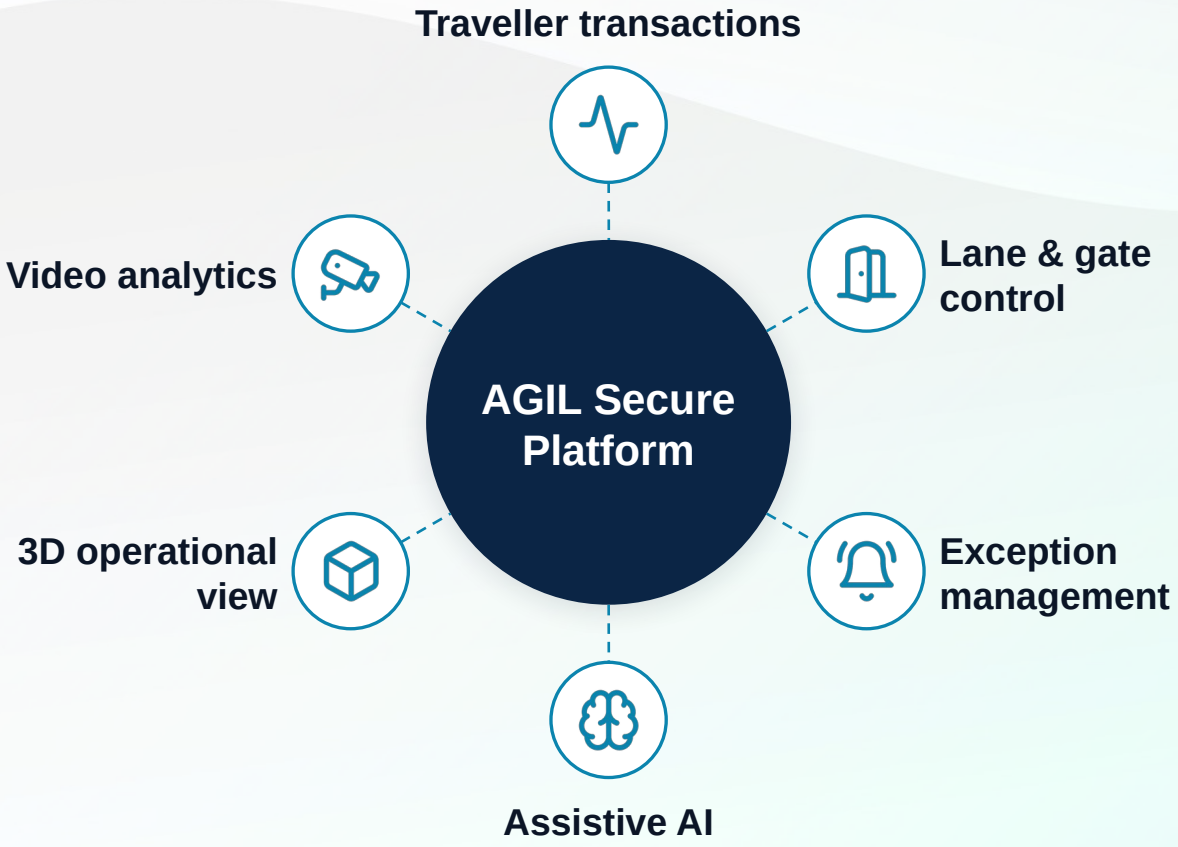
These are comps, not a valuation. No named investor has engaged with FastPass.

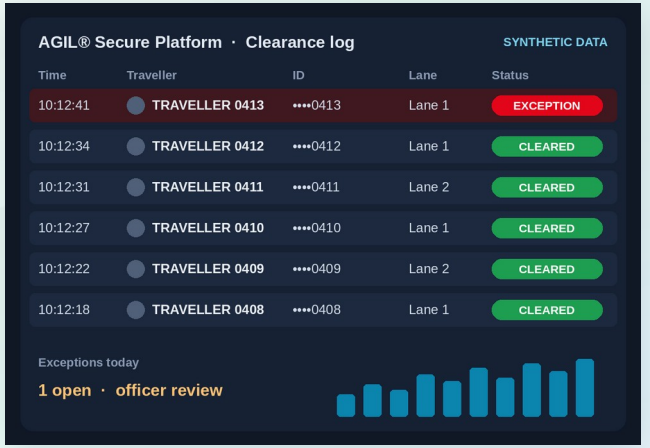
Amadeus (Apr 2024) · The Corner (EBITDA multiple) · BigBear.ai (2024) · SITA (Jul 2024) · IN Groupe (Jul 2025) · Thales (Apr 2019) · FinTech Futures (Apr 2025) · Renaissance Capital (Jun 2021)

Appendix index

AGIL® Secure Platform: the command layer behind every step.

Six capabilities. The AI recommends, the officer decides [confirm].





- Human in the loop: the AI recommends, the officer decides, every decision logged [confirm]
- Integrations: CCTV, access control, incident management [list systems, confirm]
- Third-party gates via open interfaces [confirm]: decides whether biometric-border airports are an installed base to upgrade
- Status words CLEARED / EXCEPTION; demo screens synthetic only

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Pilot design: one airport, one group, one lane.

The COO's confidence slide. The full scorecard is on A7b.

Scope

- One airport [name], one traveller group [set by the border agency], one lane, [N] months
- [Paid / co-funded]: decide before external use
- Payer and procurement route [to validate]
- Converts a lane or adds one; floor space [agree]
- Baseline on the existing lane for [N] weeks; scorecard signed by airport and agency

Fallback and officer control

- Staffed lane beside every FastPass lane; documents accepted anywhere; nobody forced into biometrics
- Covers dead phone, BLE congestion, wrong phone paired, face no-match, tailgating, outage
- Children without phones, wheelchair users, assisted travellers: [design]
- Exception → officer review → manual clearance or referral; logged

Milestone exits and trigger

- M2 exit: lane live + independent ISO/IEC 19795 and 30107-3 tests + BLE relay and mis-pairing tests → releases T3
- M3 exit: KPIs met → releases T4
- KPIs met → airport decides on [N] lanes for US$[X] within [N] months [if in the agreement]
- KPIs missed → stop or redesign; no further tranche
- Prototype to date: match [X s] · false-reject [X%] · [X/lane/hr] · BLE pairing [X%], or scripted demo

Pilot scorecard: targets set now, measured live.

Baseline, target and go / no-go agreed with the airport and agency before go-live.

Measure	Baseline	Target	Go / no-go
Throughput (travellers / lane / hr) vs measured e-gate	[measure]	[X]	[X]
Immigration seconds per traveller	[measure]	[X]	[X]
Kerb-to-gate minutes (full-rollout team target −20–30%)	[measure]	[X]	[X]
False-accept (impostor) rate	[measure]	[X]	[X]
False-reject rate	[measure]	[X]	[X]
BLE pairing, phone locked or backgrounded (iOS, Android)	[measure]	[X]	[X]
Mis-pairing rate in a crowded lane	[measure]	[X]	[X]
Exceptions per 100	[measure]	[X]	[X]
Officer minutes per 100	[measure]	[X]	[X]
Enrolment vs break-even uptake per lane [X%]	[measure]	[X]	[X]
Lane availability	[measure]	[X]	[X]
Traveller satisfaction	[measure]	[X]	[X]
Custom-engineering days (margin proxy)	[measure]	[X]	[X]

Benchmarks, not FastPass claims: e-gate 180/hr vs booth 60–70 (Riga) · Changi 25 s → 10 s · boarding −30% (SITA / Turkish trial)

AviationPros (Riga Airport / State Border Guard of Latvia) · Malay Mail (Oct 2024) · Future Travel Experience (Apr 2021)

Appendix index

Trust, privacy and security: what the CTO confirms.

Design intent, test plan and data protection. Bracketed items need CTO sign-off.

Second-factor binding [CTO]

- The lane challenges the credential on the paired phone, not just a beacon
- The live face must match that credential
- Wrong pairing → exception to an officer, never a false accept
- Where the face match runs and what crosses the BLE link [confirm]

Enrolment and testing

- Passport chip read with ICAO 9303 passive authentication, or OCR [confirm]; OCR-only is fixed before any agency pitch
- Liveness selfie; credential encrypted on the device [enclave / keystore]
- Tests [plan]: ISO/IEC 30107-3, ISO/IEC 19795, demographic accuracy
- Face algorithm [own / licensed] and NIST FRTE standing [confirm]
- BLE threats: wrong phone, relay, spoofed beacons, tailgating

Privacy and trust

- Opt-in at enrolment; retention; data controller per deployment
- PDPA (SG), PDPL (UAE, Saudi), PDPA (MY, TH), PDP Law (ID), GDPR [verify]
- Public displays: sequence number + CLEARED only
- 42% of hesitant travellers would reconsider if data protection were guaranteed (IATA GPS 2025)

Design principles: Trusted by design · Easy to adopt (SDK) · Connected ecosystem · AI-assisted decisions

IATA Global Passenger Survey 2025 · ICAO Doc 9303 · ISO/IEC 30107-3 · ISO/IEC 19795 · NIST FRTE

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How an agency could accept a phone-held credential.

Standards status, precedents and the certification plan per country.

ICAO Digital Travel Credential

- Type 1 (passport still carried): specs endorsed 2020; State implementation guide v1.0, Sep 2025
- Annex 9 SARPs proposed at FALP/14, Apr 2026: not adopted; to the Air Transport Committee, fall 2026
- Types 2 and 3: not ready

EU and US

- EES: progressive from 12 Oct 2025; fully operational 10 Apr 2026
- ACI Europe: ~70% longer processing (Commission disputes)
- EU Digital Travel app ~2030; EUDI wallets end-2026 (identity, not passports)
- US: CBP may require face biometrics from non-citizens on entry and exit (from 26 Dec 2025)
- Apple Digital ID: TSA beta, 250+ airports, domestic only

Precedents and our plan

- Finland border-guard DTC pilot: 355 checks, ~8 s; 90% would reuse
- IATA One ID wallet + biometrics proof of concept, Hong Kong–Narita (Oct 2024)
- Who approves on-the-move clearance per country: [agency, process, owner]
- Certification roadmap and launch eligibility: [per country]

ICAO (FALP/14; DTC guide) · European Commission · The Local / ACI Europe · Federal Register · Skift · Keesing · IATA (Oct 2024)

Appendix index

Risks you will raise, and the milestone that retires each.

Agency, technical, privacy, adoption, payer, market and flow risks.

Risk	Mitigation	Retired at
Agency declines the phone credential, or sees no value in a second factor	Co-design the pilot with the agency; align to ICAO DTC Type 1; Finland DTC precedent. No value → stop or redesign	M1
Accuracy, spoofing, wrong-phone pairing, relay attack	Independent ISO/IEC 19795 and 30107-3 tests; BLE mis-pairing and relay tests; face must match the paired credential	M2 exit
Privacy and data protection	Credential encrypted on the phone; opt-in; controller and retention per country [legal review]; no personal data on displays	M1
Enrolment cold start	Airline-app SDK; eligible segments; enrolment is an M3 KPI vs break-even uptake [X%]; ~3 in 4 travellers willing (IATA)	M3
Payer mismatch: airport buys, agency saves officer time	Payback split per payer and budget holder; procurement route per country; co-funding at M1	M1
Middle East 2026 traffic shock (airports −21% H1; carriers −14.2% Aug; IATA −11.4% for 2026)	SEA or dual-region pilot; sell efficiency, not growth; long-cycle builds continue (DWC 2032)	M1 market
Flow failure like EES (~70% longer, per ACI Europe; disputed)	Redesign the flow, not just the gate; staffed fallback lane	M2

ACI World (Sep 2026) · IATA (Jun, Sep 2026) · The Local / ACI Europe (Dec 2025) · Keesing (Finland DTC) · IATA GPS 2025 · Dubai Media Office · WAM (Jun 2026)

Appendix index

Risks you will raise, and the milestone that retires each.

Competitive, procurement, brand, IP and delivery risks.

Risk	Mitigation	Retired at
Incumbents win the replacement cycle first (SITA: 83% of surveyed airports biometric by 2028)	Retrofit onto installed third-party gates [CTO to confirm]; else target the replacement cycle	M1–M2
Incumbents copy the device factor	Accreditation earned in the pilot; control-room integration; [IP]	M3–M4
Procurement cycle longer than runway	Innovation-pilot route; Group channel; runway [X–Y] months; M1 date checked ([Q1 2027] is ~5 months away)	M1
Brand: Hamad / Qatar Airways / SITA 'Fast Pass' (Jul 2026)	Trademark and naming check in ME/SEA classes; fallback names ready	Before launch
Freedom to operate on the BLE lane	Patent / FTO review before any 'proprietary' claim. INTERNAL & PRIVILEGED: Axess AG US12367722B2 referred to counsel	Before launch
Matching bias · cyber attack · airline DCS integration delay	[mitigation for each]	[TBC]

Unit economics and payback per payer (placeholders).

FastPass values stay bracketed until a pilot prices them and Finance validates them.

Per lane

- Hardware US$[X] one-off
- Implementation [X%]
- Software US$[Y] / lane / yr
- SLA support US$[Z] / yr
- Gross margin per line [X%]
- Recurring share by year 3 [X%]
- Blended gross margin [X%]

Payback per payer

- Airport: added departures / lane / hr × [value per departure] + airport-staff hours redeployed, vs lane cost US$[X] = [N] months
- Agency: officer-hours redeployed [N] × loaded cost [US$/hr] = [N] months
- Stated as incremental over the existing e-gate baseline
- Where the agency gains, co-funding is part of M1

Who pays: precedents

- Aena (airport operator) ran an ABC framework
- Riga: Interior Ministry and an EU fund
- Cardiff: up to £1M from the Welsh Government (Border Force funds e-gates only above 2M inbound)
- Route per target country: [to validate]

Cost anchors, margin comparables and the 5-year plan.

Public anchors and large-company comparables: context only, not FastPass prices.

Public cost anchors

- ~€140–235k per unit (2017–18 tenders)
- Aena: EUR 90M ÷ 652 max units ≈ €138k (incl. kiosks)
- Riga: EUR 1.87M ÷ 8 gates ≈ €234k

Margin and mix (context)

- Axon: software & services ~74% GM (Q3 2025) vs 59.7% blended (FY2025)
- Evolv: subscription 56% GM, hardware bundled: a caution
- Motorola Solutions: software & services 38% of revenue

Buyer ROI context

- ~60% of airport opex is staff and contracted services (ACI FY2019; not all labour)
- ~20% of opex is security-related (ACI Europe)
- Changi 25 s → 10 s; −30% boarding (SITA / IST); ~9 min per wide-body boarding (Delta)
- Illustrative: 1 officer × 4 gates × 180/hr ≈ 720 pax/hr, vs 60–70 at a booth (not a FastPass claim)

5-year plan and demand

- Native chart once validated, labelled 'plan, internal estimate': lanes, revenue, recurring %, gross margin, burn, break-even year
- The ask derives from a bottom-up cost of M1–M4
- IATA (Mar 2026): demand more than doubles by 2050 (3.1% CAGR); supersedes '2× by 2040'

Team, Group fit and what an investor would own.

Answers the CEO ('why a venture?') and the VC ('what would I own?').

Team

- Bios: [name, role, track record, full-time status, equity] × 4 leads
- Advisors: [names]
- Hires in the programme: commercial lead (airport / border-agency sales); certification & privacy lead
- Concept to Airshow prototype: [N] months
- Drawn from Smart Security & Automation, Smart Utilities & Infrastructure and Urban Solutions

Why a venture (INTERNAL)

- [Rationale]
- Group pull-through across the three business areas; target US$[X M] by 2030

Channel conflict [TBC]

- Non-ST integrators? Airports where other ST units compete?
- Related-party sales

What an investor would own

- Structure: [NewCo spin-out or internal venture via STEV / In.Vent]
- IP and platform rights: [assigned or exclusively licensed]
- Cap table: [placeholder]; ST Engineering as [shareholder / supplier / channel]
- Group supply on arm's-length terms; no parent veto on an exit [TBC]
- If the face algorithm is licensed, say so: the value is the lane factor + control room

Every internal number, labelled and owned.

So no estimate reaches investors as an external fact.

Assumption	Label	Validation · owner
Up to 4 document checks per departure	Process framing (domestic: no immigration)	[CPO]
Up to 4 → 0 at full rollout	Design target; pilot is 1 → 0 at immigration	[CPO]
≥4 staff per checkpoint workflow	Internal estimate	Pilot baseline · [COO]
10–15% ME + SEA share of airport ABC	Internal estimate (URS)	Method [to document] · [CMO]
10–15% FastPass-serviceable share	Internal estimate (URS)	Method [to document] · [CMO]
US$10–15M a year (INTERNAL)	Candidate revenue target, ~15–23% of midpoint	Confirm in writing · [CMO / Finance]
20–30% less kerb-to-gate time	Team target, unvalidated (full rollout)	Pilot baseline · [CPO]
~78% brownfield (URS)	Unsourced; shown beside ACI's 57% (by value)	[CMO]
35+ years of secure delivery	Company claim	—
Prototype shown end to end, Feb 2026	Company record; live status per module	[CTO]
Every [X] price, margin, KPI, date, count	Placeholder until confirmed	Tiers: A16

Ten questions most likely to decide the room.

Ownership, scale, demand, payer and measured gain. Full bank of 42 in the presenter pack.

Question	Short answer
What would I own?	Not decided yet. For outside money we expect a NewCo, IP assigned or exclusively licensed, full-time leads with equity and arm's-length Group supply (A12). Until then, the first money is the Group's.
Is this venture-scale?	The US$42–94M is a deliberately narrow beachhead: ABC e-gates and kiosks in two regions. The path to US$100M+ (every touchpoint, software, other airports and borders) is a hypothesis the pilot must open.
Which agency has asked for a second factor?	None yet. That is the bet M1 tests, and if agencies don't value it we stop or redesign.
Who pays: airport or agency?	It differs by country (Aena, Riga, Cardiff precedents). We compute payback per payer and structure the pilot with the budget holder (A11).
What's your measured gain over an e-gate?	Not measured yet. The pilot measures throughput against the existing e-gate baseline, with a go / no-go agreed in advance.

Ten questions most likely to decide the room.

Demo, security, brand, market and the ask. Full bank of 42 in the presenter pack.

Question	Short answer
Was the demo real?	It is an illustrated walkthrough of the prototype demonstrated end to end at the Airshow. Real footage: [show / none]. Measured: [figures / none yet].
How is Bluetooth a second factor? Relay or mis-pairing?	Design intent: a cryptographic challenge to the paired phone's credential, plus a face match to that credential [CTO]. Independent tests are M2's exit criteria.
Can you use the name 'Fast Pass'?	Hamad, Qatar Airways and SITA launched it in Jul 2026. Our trademark review is [status], and fallback names are ready.
Why lead with the Middle East after −21% in H1 2026?	We sell efficiency, not growth; an SEA or dual-region pilot is the hedge; build-outs like DWC 2032 continue.
How much, and what's true at the end?	[US$X M] over [N] months buys M1–M4. In 12 months: a signed pilot and a live lane (M1–M2); the KPI readout (M3) follows by [Qx 2028]. Not revenue.

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The trusted identity layer for every checkpoint.

Future options, kept out of the main pitch. Airports first.

Vision: no documents shown at any checkpoint a person crosses. Airports first.

Next capabilities

- Accept airline-issued, ICAO DTC and EUDI wallet credentials as the device factor via the SDK [CTO to confirm]
- Arrivals and transit
- Lounge

Technology options

- LiDAR
- Iris
- E-concierge
- AI luggage scanning ('FastPass AI Scanning')

New markets, in order

- 1 → Congested airports worldwide
- 2 → Seaports, cruise and land borders
- 3 → Secure campuses [to confirm]

Ship gates: what must be filled, and when.

No slide reaches a room with a bracket as its focal element.

Tier 1 · before ANY showing

- Ask amount and runway; M1–M4 dates (M1 checked for realism)
- Presenter names
- Slide-6 counts and two permissioned learnings (or delete)
- Scorecard targets and go / no-go (A7)
- Per-module live confirmation, incl. BLE pairing and boarding
- Real Airshow clip or one dated photo of a person at the live gate (5c)
- Written confirmation that the app-screen faces are AI-generated (5b, slide 6, A2), else replace them
- Corporate-comms clearance of the Minister's quote

Tier 2 · before the EXTERNAL file is sent

- Structure, IP and platform owner, ST Engineering role
- Named, photographed, full-time leads
- Pilot market (SEA or dual-region); paid or co-funded
- Revenue target and year in writing, plus the bottom-up
- Trademark check on 'FastPass'; G-SRC click-check (A1)
- CTO sign-off on binding and enrolment (A8); two-factor wording tested with agencies
- Text diff, EXTERNAL vs INTERNAL

Tier 3 · can stay open

- Per-lane prices and unit economics
- 5-year plan; use of funds %
- Certification roadmap
- Channel-conflict rules
- Expansion beyond airports

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